

Comparative Evaluation and Domain Adaptation of Neural Speaker Diarization for Air Traffic Control Communications

Abstract—Speaker diarization in air traffic control (ATC) communications poses distinct challenges due to VHF radio channel effects, high speaker counts, and sparse half-duplex speech patterns. This paper presents a comparative evaluation of four neural diarization architectures on ATC radio audio: Sortformer (offline and streaming), pyannote 3.1 (clustering-based), and LS-EEND (retention-based). Using a revised ATC dataset comprising 16 recordings with 21–90 speakers per recording, we find that all systems exhibit domain-specific failure modes. Sortformer achieves the lowest diarization error rate (DER) among pretrained systems but suffers from persistent speaker confusion errors (CER), where a single speaker slot dominates the output across multiple ground-truth speakers. The streaming variant’s Arrival-Order Speaker Cache reduces CER by 20–40% relative to offline, at the cost of elevated false alarm rates. Clustering-based pyannote collapses 32 speakers to 2 due to VHF-induced embedding similarity, while LS-EEND’s telephone-trained checkpoint produces excessive false alarms on ATC noise. We demonstrate that freezing the pretrained encoder and fine-tuning only the Transformer layers on 12 ATC recordings reduces CER by 46–78% across four held-out recordings from three distinct facilities. Per-role analysis reveals preferential improvement on controller speech (45–67% CER reduction) over pilot speech (37–48%), and speaker embedding analysis confirms that an alternative embedding architecture maintains meaningful speaker separability on ATC audio where the clustering pipeline’s embeddings collapse. These findings establish Sortformer as the most suitable baseline for ATC diarization and demonstrate that lightweight domain adaptation yields substantial and generalizable performance gains.

Index Terms—speaker diarization, air traffic control, domain adaptation, VHF radio, speaker confusion, neural diarization

I. INTRODUCTION

Air traffic control (ATC) communications rely on voice exchanges between controllers and pilots over Very High Frequency (VHF) radio channels. As ATC workloads increase [1], automated speech processing systems that can extract structured information from these communications—including speaker identity, clearance content, and compliance verification—have become an active area of research [2], [3]. Speaker diarization, the task of determining “who spoke when,” is a critical upstream component for such systems: misattributing a clearance to the wrong speaker can lead to incorrect downstream processing in safety-critical contexts.

However, ATC radio audio presents several challenges that distinguish it from the conversational telephone or meeting speech on which modern diarization systems are typically trained and evaluated. The VHF amplitude-modulated radio channel restricts audio bandwidth to approx-

imately 300–3400 Hz, stripping away high-frequency speaker-discriminative features that are available in wideband audio [4]. Automatic squelch gating introduces abrupt silence-to-speech transitions. Controller-pilot exchanges are half-duplex (push-to-talk), producing sparse speech patterns with long silence gaps between transmissions—very different from the near-continuous overlapping speech in meeting or telephone conversations. Additionally, a single ATC frequency may carry transmissions from dozens of distinct speakers (21–90 in our dataset), far exceeding the 2–6 speakers typical of standard diarization benchmarks [5], [6].

Recent end-to-end neural diarization systems have achieved strong results on standard benchmarks. Sortformer [7] uses Sort Loss to resolve speaker permutation in arrival-time order, achieving competitive DER on CALLHOME and DIHARD III. Its streaming extension [8] introduces an Arrival-Order Speaker Cache (AOSC) for real-time speaker tracking. LS-EEND [9] employs retention mechanisms for linear-complexity streaming diarization. Clustering-based pipelines such as pyannote [10] remain widely used, combining neural segmentation with agglomerative clustering of speaker embeddings. However, none of these systems have been systematically evaluated on ATC radio audio, and their failure modes in this domain remain unexplored.

In this paper, we make the following contributions:

- 1) We present the first systematic comparison of four neural diarization architectures on ATC radio audio, identifying architecture-specific failure modes for each system.
- 2) We demonstrate that Sortformer’s speaker confusion error on ATC audio is reduced by its streaming variant’s AOSC mechanism (20–40% relative CER reduction), establishing a concrete benefit of chunked processing with speaker caching.
- 3) We show that fine-tuning only the Transformer layers of Sortformer on ATC data yields 46–78% CER reduction across recordings from three distinct ATC facilities, including two facilities absent from the training data, confirming cross-facility generalization.
- 4) We provide a speaker embedding analysis showing that an alternative embedding extractor maintains meaningful speaker separability (cosine similarity margin 0.28–0.43) on ATC audio where the clustering pipeline’s embeddings collapse, demonstrating that the clustering failure is architecture-specific rather than domain-inherent.

II. RELATED WORK

A. End-to-End Speaker Diarization

End-to-end neural diarization (EEND) was introduced by Fujita et al. [11], formulating diarization as frame-wise multi-label classification with permutation-free objectives. Subsequent work extended EEND with encoder-decoder attractors [12] for handling unknown speaker counts and self-attention mechanisms [13] for improved temporal modeling.

Sortformer [7] departs from permutation-invariant training by introducing Sort Loss, which trains the model to predict speakers in arrival-time order. Built on a self-supervised pre-trained Fast Conformer encoder [14] with an 18-layer Transformer, Sortformer achieves state-of-the-art results on multiple benchmarks. Its streaming extension [8] introduces the AOSC, which stores frame-level acoustic embeddings of previously observed speakers, enabling real-time diarization with arrival-time ordering at latencies as low as 0.32 s. A recent benchmarking study [15] noted that all Sortformer versions exhibit elevated speaker confusion on VoxConverse, which features higher average speaker counts, suggesting out-of-distribution generalization gaps.

LS-EEND [9] replaces standard self-attention with a retention mechanism for linear temporal complexity, enabling frame-synchronous streaming diarization of hour-long recordings. Its online attractor decoder maintains per-speaker attractors that are updated frame-by-frame, providing continuous self-correction.

B. Clustering-Based Diarization

The pyannote pipeline [10], [16] combines a neural segmentation model with Emphasized Channel Attention, Propagation and Aggregation TDNN (ECAPA-TDNN) [17] speaker embeddings and agglomerative clustering. While effective on wideband conversational audio, clustering approaches depend critically on the quality of speaker embeddings. VBx [18] and DiariZen [19] have pushed clustering-based methods further by incorporating self-supervised representations.

C. ATC-Specific Diarization

Han et al. [4] proposed a system specifically designed for ATC radio communications that combines voice activity detection, end-to-end speaker separation, and probabilistic knowledge-based text clustering. Their work explicitly acknowledges that VHF radio processing causes “voiceprint blurring,” making acoustic-only diarization unreliable, and introduces text-based clustering as a complementary approach. Zuluaga-Gomez et al. [3] demonstrated speaker role detection from ASR transcripts using fine-tuned BERT, achieving 0.90/0.95 F1 on controller/pilot role classification.

III. EXPERIMENTAL SETUP

A. Dataset

We use a revised version of a publicly available ATC corpus [20] with corrected timestamps, casing, punctuation, and four-level audio quality annotations. The dataset comprises

TABLE I
EVALUATION RECORDING CHARACTERISTICS

ID	Facility	Sector	Spk.	Dur.
R1	DCA	Departure	32	88 min
R2	DCA	Departure	31	60 min
R3	DFW	Arrival	43	65 min
R4	LOG	En-route	88	120 min

16 recordings totaling approximately 22 hours of ATC radio audio from three U.S. ATC facilities, which we refer to as Facility A (12 recordings), Facility B (2 recordings), and Facility C (2 recordings). Speaker counts per recording range from 21 to 90 unique speakers. All audio is single-channel VHF radio at the receiver, capturing both controller and pilot transmissions on the same frequency.

Ground truth speaker annotations are derived from time-stamped transcription files. Each annotation specifies a speaker identifier, listener, start/end times, and transcript. Speaker identifiers follow ATC conventions: controllers are identified by position codes (e.g., containing facility and sector designators), while pilots use airline callsigns. Unintelligible segments (lowest quality level) are excluded from evaluation. Reference files in Rich Transcription Time Marked (RTTM) format are generated from these annotations.

Table I summarizes the four evaluation recordings used throughout this paper.

B. Systems Under Evaluation

We evaluate four diarization systems representing three architectural paradigms:

Sortformer Offline: A 123M-parameter end-to-end model with a self-supervised pre-trained Fast Conformer encoder and 18-layer Transformer [7], producing 4-speaker frame-level probabilities. Inference uses 10 s windows with 5 s shift and bfloat16 mixed precision.

Streaming Sortformer: The streaming variant [8] with AOSC, evaluated at 10 s latency. Uses the same windowed evaluation protocol as offline for comparability.

pyannote 3.1: Three-stage clustering pipeline using neural segmentation, ECAPA-TDNN speaker embeddings, and agglomerative clustering [10]. Processes full recordings without windowing.

LS-EEND: A 11.2M-parameter model with Conformer encoder and Multi-Scale Retention decoder [9]. We ported the original codebase from PyTorch 1.13 to PyTorch 2.6, addressing four compatibility issues. The CALLHOME-finetuned checkpoint is used with feature normalization to compensate for the ATC-telephone feature distribution gap (ATC log-mel std ≈ 0.31 vs. CALLHOME std ≈ 1.0).

C. Evaluation Protocol

DER is computed with a 0.25 s collar and without ignoring overlapping speech, following the strictest evaluation convention [5]. DER decomposes into three components: false alarm (FA, speech detected where none exists), missed speech

TABLE II
DER (%) AND COMPONENTS FOR PRETRAINED SYSTEMS

System	Rec.	DER	FA	MISS	CER
Sortformer (offline)	R1	20.9	2.7	1.4	16.8
	R2	20.2	2.4	0.7	17.1
	R3	47.4	31.0	0.5	15.9
	R4	44.7	33.1	1.5	10.1
Sortformer (streaming)	R1	34.8	27.6	0.5	6.8
	R2	43.3	33.2	0.3	9.8
pyannote 3.1	R1	32.5	—	—	—
LS-EEND	R1	43.6	2.5	10.2	30.9
	R2	49.1	2.6	5.6	40.8

(MISS, speech present but not detected), and speaker confusion error rate (CER, correctly detected speech attributed to the wrong speaker).

IV. COMPARATIVE EVALUATION RESULTS

Table II presents DER and its components for all four systems on the evaluation recordings. Sortformer offline achieves the lowest DER on Facility A recordings (20.2–20.9%) but exhibits elevated DER on Facility B and C recordings (44.7–47.4%), primarily due to high false alarm rates. Across all recordings, speaker confusion (CER) is the dominant error component for Sortformer, consistent with a pattern we term “lock-up”: a single speaker output slot dominates the model’s predictions regardless of the active ground-truth speaker.

A. System-Specific Failure Modes

Each system fails on ATC audio through a distinct mechanism:

Sortformer (offline): Speaker confusion dominates (CER 10.1–17.1%). Visual analysis of per-window probability outputs reveals a persistent pattern where one speaker slot saturates near 1.0 across multiple consecutive ground-truth speaker transitions, while other slots remain suppressed. This lock-up pattern indicates that the model correctly detects speech activity but assigns it to a single speaker identity.

Sortformer (streaming): The AOSC reduces CER substantially (6.8% vs. 16.8% on R1, 9.8% vs. 17.1% on R2), but false alarm rates increase dramatically (27.6–33.2% vs. 2.4–2.7%). The streaming model produces systematically elevated speaker probabilities during non-speech regions, because it was trained on conversational audio with near-continuous speech. Threshold calibration (0.72 instead of 0.50) partially mitigates this, reducing DER from 34.8% to 21.4% on R1.

pyannote 3.1: Catastrophic under-clustering occurs, with 32 ground-truth speakers collapsed to 2 detected speakers on R1 and 46 speakers to 1 on another recording. The ECAPA-TDNN speaker embeddings lose discriminative power on narrow-band VHF audio.

LS-EEND: The CALLHOME-finetuned checkpoint produces high CER (30.9–40.8%) because it was optimized for 2–6 speaker telephone conversations. Despite feature normalization, the model collapses ATC’s many speakers into 1–2

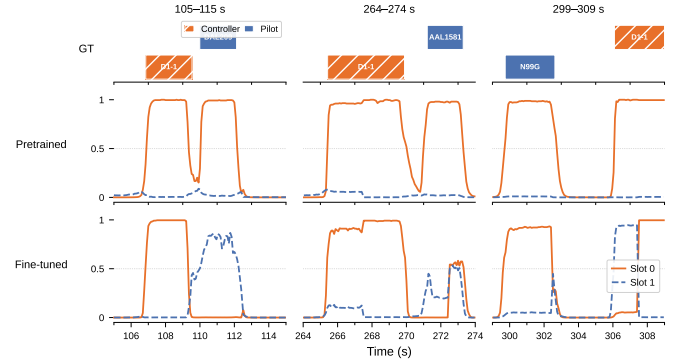


Fig. 1. Lock-up pattern and fine-tuning correction on three controller–pilot exchanges from R1. Top row: ground truth (orange hatched = controller, blue = pilot). Middle row: pretrained Sortformer—Slot 0 dominates both speakers. Bottom row: fine-tuned—Slot 1 activates for pilot transmissions.

attractor slots. The retention mechanism avoids the lock-up pattern observed in Sortformer but cannot compensate for the training domain mismatch.

V. ANALYSIS OF SPEAKER CONFUSION

A. Visual Analysis of Lock-Up Pattern

Fig. 1 illustrates the lock-up pattern and its correction through fine-tuning across three controller–pilot exchanges from recording R1. Each column shows the ground truth (top), pretrained Sortformer output (middle), and fine-tuned output (bottom) for a 10 s window containing a controller transmission followed by a pilot response.

In all three pretrained panels, Slot 0 (solid line) activates for both the controller and the pilot, assigning both speakers to the same output slot. Slot 1 (dashed line) remains near zero throughout. This is the lock-up pattern: the model detects speech correctly but fails to distinguish between speakers, defaulting to a single dominant slot.

After fine-tuning, the behavior changes: Slot 0 activates for the controller while Slot 1 rises for the pilot transmission. The consistency of this correction across three temporally separated exchanges (at 105 s, 264 s, and 299 s) confirms that fine-tuning produces a systematic improvement in speaker discrimination rather than a localized artifact.

B. Attention Entropy Observations

We extracted attention weight matrices from all 18 Transformer layers during inference using forward hooks, computing per-layer Shannon entropy $H = -\sum_i a_i \log a_i$ averaged across 8 attention heads and all query positions (maximum uniform entropy: $\ln(125) = 4.83$ nats for 125-frame windows).

On R1, a window with correct speaker tracking maintains entropy of 4.68 nats at layer 17, while a window exhibiting lock-up drops to 3.98 nats—a gap of 0.70 nats (Fig. 2). However, this relationship is not consistent across recordings: on R2, the high-CER window showed *higher* entropy (4.26 nats) than the low-CER window (3.97 nats), and per-window entropy-CER correlation was not statistically significant ($r = 0.076$, $p = 0.789$). We conclude that attention

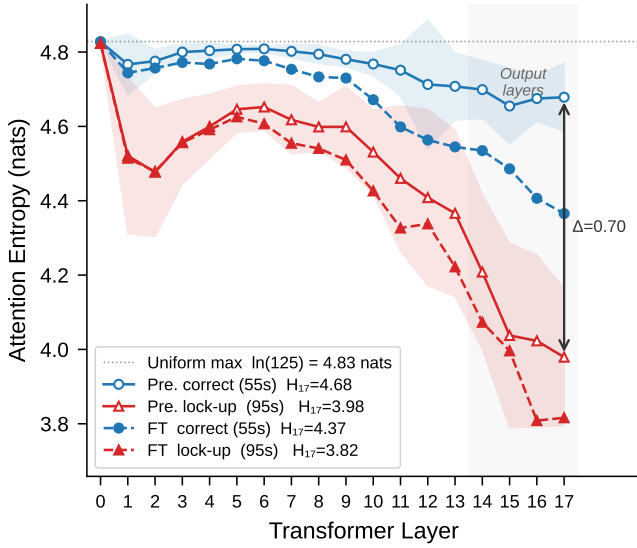


Fig. 2. Layer-wise attention entropy for pretrained (solid) and fine-tuned (dashed) Sortformer on R1. Circle markers: correct-tracking window; triangle markers: lock-up window. Shaded bands show $\pm 1\sigma$ across heads. Fine-tuning reduces entropy in both windows but narrows the correct/lock-up gap from 0.70 to 0.55 nats.

concentration is a feature of both correct and incorrect speaker assignment, and that entropy alone is not a reliable predictor of confusion errors.

C. Speaker Embedding Analysis

To determine whether pyannote’s under-clustering failure reflects a fundamental limitation of speaker embeddings on ATC audio, we extracted TitaNet-Large [21] embeddings from individual speaker cues in R1 (32 speakers, 389 cues) and R4 (88 speakers) at both 16 kHz and 8 kHz (band-limited to match VHF bandwidth).

Fig. 3 shows the intra-speaker and inter-speaker cosine similarity distributions for R1 at 16 kHz. The separability margin (mean intra-speaker minus mean inter-speaker similarity) is 0.318 for 32 speakers and 0.429 for 88 speakers on R4. Band-limiting to 8 kHz reduces the margin by only 0.02–0.04, confirming that the VHF channel’s native bandwidth limitation dominates over sample-rate effects. TitaNet maintains meaningful separation where pyannote’s ECAPA-TDNN collapses, indicating that the clustering failure is specific to the embedding architecture rather than an inherent property of ATC audio.

VI. DOMAIN ADAPTATION VIA FINE-TUNING

Based on the comparative evaluation, we select Sortformer as the most promising system for ATC diarization and investigate whether lightweight domain adaptation can reduce the speaker confusion error.

A. Fine-Tuning Setup

We freeze the 115M-parameter pre-trained encoder and fine-tune only the 18-layer Transformer and output heads

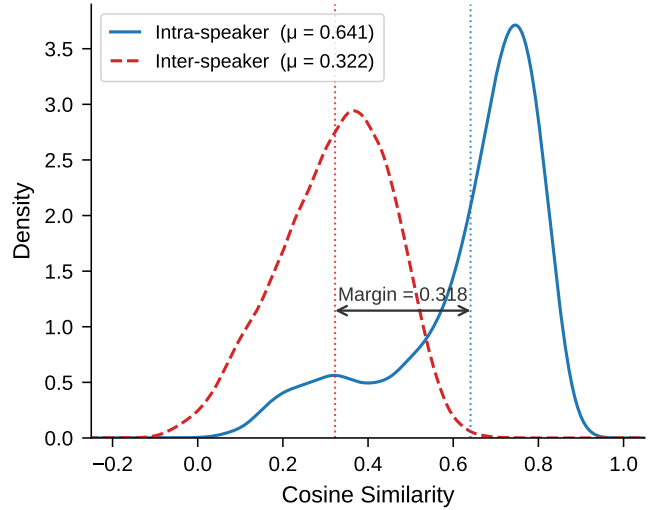


Fig. 3. Speaker embedding similarity distributions for R1 (32 speakers, 16 kHz). Solid: intra-speaker; dashed: inter-speaker. Separability margin of 0.318 indicates meaningful speaker discrimination on VHF radio audio.

TABLE III
EFFECT OF ATC FINE-TUNING ON DER COMPONENTS (%)

	Rec.	Fac.	DER	FA	MISS	CER
Pre.	R1	A	20.9	2.7	1.4	16.8
	R2	A	20.2	2.4	0.7	17.1
	R3	B	47.4	31.0	0.5	15.9
	R4	C	44.7	33.1	1.5	10.1
F.T.	R1	A	12.6	1.6	1.9	9.1
	R2	A	10.9	1.6	1.6	3.8
	R3	B	21.2	13.1	2.0	6.1
	R4	C	11.8	4.3	2.1	5.4

(8.15M trainable parameters) on ATC data. Training uses 12 recordings from Facility A (1,453 windows of 90 s with 45 s shift), with four recordings from three facilities held out for evaluation (Table I). We use AdamW with $\text{lr} = 10^{-5}$ ($10\times$ lower than pretraining), 100-step warmup, and train for 1,000 steps with batch size 4 and bf16 precision.

B. Results

Table III shows DER before and after fine-tuning. CER decreases by 46–78% across all four evaluation recordings. The improvement generalizes across facilities: R3 (Facility B, absent from training data) achieves 62% CER reduction, and R4 (Facility C, absent) achieves 47%.

Fig. 4 visualizes the DER decomposition before and after fine-tuning. The reduction is primarily in CER (dark segments), while FA decreases substantially on Facility B and C recordings. MISS increases slightly (from $\sim 1\%$ to $\sim 2\%$), indicating a modest trade-off where the model becomes slightly more conservative about activating speaker slots.

C. Per-Role Analysis

Since ATC communications involve two distinct speaker roles (controllers and pilots), we analyze CER separately

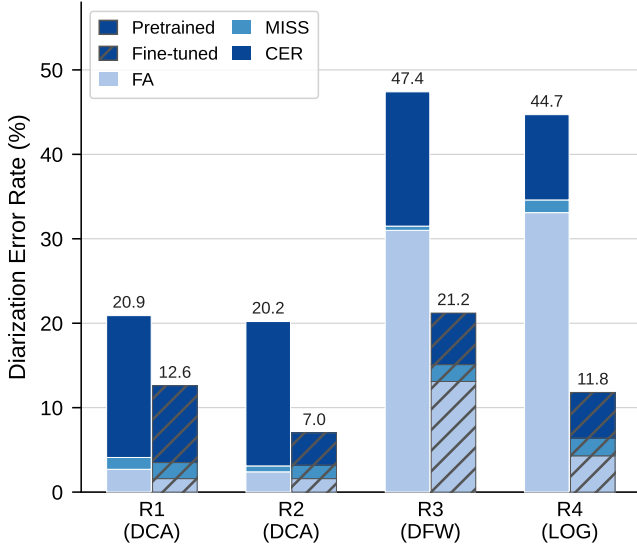


Fig. 4. DER decomposition before (solid fill) and after fine-tuning (hatched) across four evaluation recordings from three facilities. Light: FA; medium: MISS; dark: CER. Fine-tuning reduces CER by 46–78% and substantially reduces FA on Facility B and C recordings.

by role. Controllers speak 55–63% of the time and exhibit more consistent acoustic characteristics within a recording. Fine-tuning reduces controller CER by 45–67% compared to 37–48% for pilot CER. This differential improvement likely reflects the greater within-session consistency of controller speech, allowing the model to learn controller-specific acoustic patterns more readily.

D. Effect on Attention Patterns

Fig. 2 compares attention entropy before and after fine-tuning. On the R1 lock-up window, fine-tuning reduces entropy from 3.98 to 3.82 nats, while the correct-tracking window drops from 4.68 to 4.37. The entropy gap narrows from 0.70 to 0.55 nats. This indicates that fine-tuning does not prevent attention concentration but rather redirects it: the model uses more focused attention overall, but the focus is better aligned with speaker-discriminative features in ATC audio.

VII. DISCUSSION

A. Domain Mismatch as the Primary Challenge

The central finding of this work is that ATC diarization failure is primarily a domain mismatch problem rather than an architectural limitation of any single system. All four evaluated systems—spanning end-to-end, clustering-based, and retention-based architectures—fail on ATC audio through different mechanisms, but all failures trace back to the mismatch between their training domains and ATC radio characteristics.

Sortformer’s advantage over the other systems comes not from an inherently superior architecture for ATC but from the diversity of its pretraining data: the self-supervised encoder

was trained on a large mixture of speech domains, providing more robust acoustic representations than the embedding extractors or encoders used by the other systems.

B. Streaming AOSC as a Partial Mitigation

The consistent CER reduction from streaming Sortformer’s AOSC (20–40% relative) suggests that chunked processing with speaker caching provides meaningful self-correction against the lock-up pattern. The AOSC stores representative embeddings for previously observed speakers, preventing the model from converging to a single speaker interpretation across the entire window. For real-time ATC information extraction, the streaming variant is required regardless of latency constraints, and its lower CER is a net benefit even accounting for the elevated FA (which can be mitigated through threshold calibration).

C. Limitations

Several limitations should be noted. First, the dataset uses single-channel audio from VHF receivers; multi-channel recordings would enable spatial diversity. Second, the Sortformer model is limited to 4 simultaneous speaker output slots; while ATC’s half-duplex nature means at most 1–2 speakers are active at any instant, a higher slot count might improve tracking of rapid speaker alternation. Third, the LS-EEND comparison uses a checkpoint trained on telephone speech, which is not a fair architectural comparison; training LS-EEND on the same ATC data would be needed to isolate architectural effects from training data effects. Fourth, we did not evaluate text-based diarization approaches [3], [4], which exploit ATC phraseology patterns and may complement acoustic diarization.

D. Future Directions

Three directions emerge from this work. First, hybrid acoustic-text diarization, combining Sortformer’s frame-level predictions with callsign detection from ASR transcripts, could address the remaining confusion errors by leveraging the structured nature of ATC phraseology. Second, extending Sortformer’s output capacity beyond 4 speakers would better match ATC’s high speaker-count environment. Third, the strong cross-facility generalization observed after fine-tuning (Facility B: –62% CER, Facility C: –47%) suggests that a modest amount of labeled ATC data from diverse facilities could produce a robust general-purpose ATC diarization model.

VIII. CONCLUSION

We presented the first systematic comparison of neural speaker diarization systems on ATC radio audio. Our evaluation of Sortformer (offline and streaming), pyannote 3.1, and LS-EEND reveals that each system fails through a distinct domain-specific mechanism: speaker confusion in Sortformer, embedding collapse in pyannote, and training-domain mismatch in LS-EEND. Fine-tuning Sortformer’s Transformer layers on 12 ATC recordings while freezing the encoder

reduces speaker confusion by 46–78% across four held-out recordings from three facilities. The streaming AOSC provides an additional 20–40% CER reduction. These results establish a practical path toward reliable speaker diarization for real-time ATC information extraction systems.

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